EmPath Project Roadmap

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EmPath is a Python-based GeoAI prototype comparing residential access to temples, groceries, and pharmacies through environmental walking friction and distance. The study covers Plano ZIP areas 75025, 75024, 75023, and 75074. Development is limited to one week and approximately 10 hours, primarily by one developer.

Data (2 hours). Verify coverage, document sources, and clean essential inputs. Deliverables: sources and coverage table and usable datasets.

Network (2 hours). Connect residential origins and destinations to a pedestrian graph; inspect crossings and connectivity. Deliverables: walking graph and map.

GeoAI and friction (2 hours). Adapt existing models or methods to derive environmental features or costs. Separate model outputs from assumed penalties. Deliverables: documented GeoAI component and weighted graph.

Analysis (1.5 hours). Calculate routes and residential rankings; compare distance-only results and audit sample routes. Deliverables: comparisons, rankings, and limitations.

Demonstration (1.5 hours). Explain results through one interactive map. Deliverable: runnable prototype.

Handoff (1 hour). Document execution and prepare a five-minute presentation. Deliverables: README, references, slides, and backup screenshots.

The minimum prototype represents all four areas using sample origins, three destination categories, one environmental scenario, and one meaningful GeoAI component. Model outputs require appropriate validation; canopy is a shade proxy, and accessibility scores do not guarantee safety. An optional teammate can review data, networks, or models without introducing an

ArcGIS Pro dependency. Future expansion would support lowest-friction navigation between selected locations.